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RESEARCH ARTICLE

Latent Spatial Regimes in Spatio-Temporal State-Space Models: Evidence on Local Price Leadership in Italian Fuel Markets

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Abstract

Data indexed in space and time, such as measurements at monitoring stations or prices at retail outlets, are often described by hierarchical state-space models, with a few latent temporal processes and a spatially correlated error. Most models assume that every parameter is common to all locations: they describe spatial dependence, but not spatial heterogeneity. We introduce the spatially clustered STEM (SC-STEM) model, which extends spatially clustered regression to a spatio-temporal geostatistical design. It partitions the locations into a few latent spatial regimes, each with its own regression coefficients, latent dynamics and spatial covariance. The partition and the parameters are estimated jointly by maximizing a likelihood with a Potts-type penalty, which favors spatially cohesive regimes without imposing them; a parametric bootstrap that re-estimates the partition carries its uncertainty into every interval and test. In simulations, the procedure rarely splits a homogeneous network, recovers well-separated regimes from one hundred locations, and, with the right number of regimes, gives estimates centered on the true values. We use SC-STEM to study local predictive price leadership in the fuel industry of eleven Italian cities with Granger-causality tests on weekly prices of 3,136 stations: the past prices of the nearest competitor help predict the price of a station in most regimes, the regimes follow brands rather than city districts, a model without regimes misses this leadership, and the 2022 excise duty cut reached consumers at different speeds within the same city. The approach applies to any network of georeferenced series whose behavior may change across space.

KEYWORDS:

Spatio-temporal models, Spatial regimes and heterogeneity, EM algorithm, Potts penalty, Retail fuel prices

1 | INTRODUCTION

Many data sets are indexed in space and in time, that is, observations come from a fixed set of locations, each observed repeatedly over time. Environmental monitoring stations and sensors produce such data, and so do retail outlets, firms and fuel stations. Such data are increasingly common in industry and in data science, where sensor networks, retail chains and digital platforms record them routinely and at a fine spatial scale. Hierarchical models describe them through a few latent processes that carry the temporal dynamics, a structure that loads these processes on the locations, and a residual variance split between a local component and a spatially correlated one[1].

In state-space form, these models are often estimated by maximum likelihood with the expectation-maximization (EM) algorithm[2], the Kalman filter and smoother supplying the moments of the latent states[3,4]. The spatio-temporal expectation-maximization (STEM) model[5,6,7] belongs to this class: it combines a regression on observed covariates, a latent vector autoregression shared by the whole network, and a measurement error with a nugget and an exponentially decaying spatial correlation. The model is parsimonious and interpretable, and its EM algorithm has closed-form steps for all the parameters except the two of the spatial covariance. It has been widely used in environmental statistics[8].

STEM naturally encodes *spatial dependence*: nearby locations have correlated residuals, and the correlation decays with distance, as Tobler's first law of geography states[9]. However, it does not encode *spatial heterogeneity*, that is, the fact that the relationships themselves may change from place to place[10,11]. One coefficient vector, one transition matrix, one nugget, one partial sill and one range serve the whole network. Two locations may be correlated, but they cannot obey different laws.

This is a strong assumption in many applications. The effect of a covariate may depend on local conditions that the data do not record. Shocks may persist longer in some areas than in others. The split of the residual variance between a local and a spatially structured component may differ between dense and sparse parts of the network. Retail industries offer a clear example: in a city, some fuel stations follow the prices of their nearest competitors closely, while others set their prices with little regard for them, depending on brand strategies and local competition. A single set of parameters averages these behaviors, and the contrast of interest is lost.

The literature offers two main answers to spatial heterogeneity. The first keeps one model and lets its parameters vary smoothly over space, as in geographically weighted regression[12,13], spatially varying coefficient models[14,15] and non-stationary covariance functions[16,17]. The second uses a mixture, in which several regimes coexist and each observation belongs to them with some probability[18,19]. Both work well when the variation they assume matches the data. Both are less natural when heterogeneity is *regime-like*, that is, when the locations split into a few groups that are homogeneous inside and different from each other, and the groups are themselves of interest[20,21].

1.1 | A spatially clustered route to spatial regimes

We take a third route and extend spatially clustered regression to a spatio-temporal geostatistical design, in which point-referenced locations are each observed over time. Clusterwise regression[22,23] assumes that the units belong to latent clusters, each with its own coefficients, and estimates the partition together with the parameters. Sugasawa and Murakami[24] made the idea spatial with a Potts-type penalty[25] on a neighborhood graph, which encourages, but does not force, spatially contiguous clusters. Within this framework, we propose the spatially clustered STEM (SC-STEM) model for geostatistical data. SC-STEM partitions the locations into a small number of latent regimes and gives each regime a complete parameter set: its own regression coefficients, its own latent dynamics, and its own nugget, partial sill and range. Regimes may thus differ in level, in their response to the covariates, in their behavior over time and in the spatial organization of their residuals. The model lies between a pooled (i.e., single cluster) model, which imposes homogeneity, and separate models for each location, which give up borrowing strength, and the spatial penalty controls where the estimate falls between the two.

SC-STEM belongs to the literature on grouped heterogeneity in panel data[26,27], in which the units may differ, but only across a few latent groups, whose members share one relation and pool their information to estimate it. The approach has since been extended to discretize continuous heterogeneity[28], to quantile regression[29], to structural breaks that differ across groups[30], to spatial dynamic panels[31] and to spatially constrained partitions[32]. The methodology has found fertile ground in socioeconomic and environmental applications, such as crime analysis[29], wages and labor market participation[28], income and democracy[30] and the relationship between emissions and growth[33]. SC-STEM brings grouped heterogeneity to spatio-temporal state-space models, and departs from these panel models in three respects. First, each group carries a complete spatio-temporal model, with its own latent dynamics and its own spatial covariance of the errors, rather than its own slopes or time effects, with spatial dependence, when present, through a spatial lag of the response[31]. Second, the spatial cohesion of the groups is encouraged by a penalty that the data may accept or decline, as a Markov random field prior does in Pan et al.[32]. Third, the uncertainty of the estimated groups enters inference through the bootstrap, rather than being set aside by asymptotic arguments under which, as the series lengthen, the estimated memberships coincide with the true ones.

So far, the spatially clustered regression device has been extended to spatial autoregressive models[34], to panel data with group fixed effects[35] and to small area estimation with the Fay–Herriot model[36]. However, several specificities of the spatio-temporal design call for specific remarks. The linear spatially clustered regression[24] and the spatially clustered Fay–Herriot model[36] are cross-sectional: given the parameters of its cluster, a unit is independent of the others, and space enters only through the penalty. The spatially clustered spatial autoregressive model[34] links the units through a spatial lag of the response, rather than through a spatial covariance of the errors, and has no temporal dynamics. The model of Cerqueti et al.[35] is, like STEM, a model for panel data, but its groups enter through fixed effects, with no temporal dynamics and no spatial covariance of their own. SC-STEM is instead a hierarchical state-space model where each regime has its own latent dynamic process, estimated by the Kalman filter,

and its own measurement error with a continuous spatial covariance. What SC-STEM shares with these models is the estimation principle, that is, a likelihood penalized by a Potts term, maximized by alternating between the parameters and the labels, with the number of clusters and the penalty chosen by information criteria. However, the spatio-temporal structure changes each of these elements: the likelihood of a regime does not factorize over its locations, so the label of a location is scored by a pseudo-likelihood; each location contributes a whole series, so the penalty needs a data-driven scale; and the bootstrap must reproduce the latent dynamics and the spatial covariance of every regime.

1.2 | Organization of the paper

The remainder of the paper is organized as follows. Section 2 defines the SC-STEM model, its objective and its estimation algorithm. Section 3 presents the computational, algorithmic and inferential aspects that matter for practical and interpretative purposes. Section 4 reports a simulation study, which shows that the procedure can be trusted where it matters: it rarely splits a homogeneous network, it recovers well-separated regimes from networks of moderate size, and it needs long series or many locations to separate weak regimes. Section 5 uses SC-STEM to study local price leadership among the fuel stations of the main Italian cities. Among others, our application of SC-STEM provides three relevant insights for competition authorities and fiscal policymakers: the price of a station follows that of its nearest competitor in most regimes; regimes reflect brand pricing strategies rather than districts; a model without regimes would miss local leadership where the regimes differ. Finally, Section 6 concludes. Three supplements complete the paper: Computational and Algorithmic Supplement A gives the full extent of the computational and algorithmic details, with their derivations and numerical experiments; Simulation Supplement B describes the simulation design and reports the extended results; and Application Supplement C describes the data of the application in detail and reports the estimates of every model and every regime.

2 | THE SPATIALLY CLUSTERED STEM MODEL

2.1 | The model

Let z_{it} be the response variable observed at location s_i (with $i = 1, \dots, d$) and time $t = 1, \dots, T$, and let $x_{it} \in \mathbb{R}^r$ be the covariates observed there. The locations are partitioned into K latent regimes, $\mathcal{P} = \{C_1, \dots, C_K\}$, and $k_i \in \{1, \dots, K\}$ denotes the regime

of location i. For a location i in regime k, the SC-STEM model is given by the system

$$z_{it} = x_{it}^{\top}\beta_k + A_i\, y_t^{(k)} + e_{it}, \qquad e_t^{(k)} \sim N\left(0, \Sigma_{e,k}\right), \tag{1}$$

$$y_t^{(k)} = G_k\, y_{t-1}^{(k)} + \eta_t^{(k)}, \qquad \eta_t^{(k)} \sim N\left(0, \Sigma_{\eta k}\right), \tag{2}$$

$$y_0^{(k)} \sim N\left(m_{0k}, C_0\right), \tag{3}$$

$$\Sigma_{e,k} = \sigma_{\epsilon k}^2 I_{d_k} + \sigma_{\omega k}^2\, C_{\theta_k}, \qquad [C_{\theta_k}]_{ij} = \exp\{-\theta_k h_{ij}\}, \tag{4}$$

where $d_k = |C_k|$, h_{ij} is the distance between locations i and j, $e_t^{(k)}$ collects the measurement errors of the locations of regime k, and A_i is the i-th row of a known $d \times p$ loading matrix. The latent states $y_t^{(k)} \in \mathbb{R}^p$, the innovations and the errors are mutually independent, within and across regimes. The parameters of regime k are

$$\Psi_k = \left(\beta_k,\ \sigma_{\epsilon k}^2,\ \sigma_{\omega k}^2,\ \theta_k,\ G_k,\ \Sigma_{\eta k},\ m_{0k}\right), \qquad k = 1, \ldots, K, \tag{5}$$

with the initial variance C_0 fixed. The model is univariate in the response and Gaussian: z_{it} is a scalar, the same variable measured at every location, and the errors, the innovations and the initial state are normal. The transition matrix G_k and the innovation covariance $\Sigma_{\eta k}$ are diagonal by default, and the locations are fixed over time: the network does not change.

With $K = 1$ the model is the STEM model of Fassò and Cameletti[5,7]. SC-STEM generalizes it by a spatially clustered approach: every component of the parameter set may change across regimes, and the regimes are estimated from the data. The pooled STEM fit is therefore the natural reference against which any estimated partition has to justify itself.

Each component has a clear reading. The coefficients β_k give the effect of the covariates in regime k. The latent state $y_t^{(k)}$ carries the temporal dynamics that the covariates do not explain and that the locations of the regime share; we use $p = 1$ and $A = \mathbf{1}_d$, that is, a single factor loading equally on every location. The transition coefficient G_k measures the persistence of these dynamics (i.e., a shock taking about $1/(1-G_k)$ periods to fade), and $\Sigma_{\eta k}$ is the variance of their innovations. In the measurement error, $\sigma_{\epsilon k}^2$ is the nugget, the variability of a single location that no neighbor shares; $\sigma_{\omega k}^2$ is the partial sill, the spatially structured variance; and $1/\theta_k$ is the distance over which the spatial correlation decays. Regimes may thus differ not only in level and in their response to the covariates, but also in their dynamics and in the spatial organization of their residuals. The partition is a parameter, and the boundaries between regimes are an output of the analysis.

Given the partition, each regime is a STEM model on its own locations. Its log-likelihood follows from the prediction-error decomposition of the Kalman filter[3,4],

$$\ell_k(\Psi_k) \;=\; -\frac{1}{2}\sum_{t=1}^{T}\Big\{ d_{kt}\log(2\pi) + \log|F_{kt}| + v_{kt}^{\top} F_{kt}^{-1} v_{kt} \Big\}, \tag{6}$$

where v_{kt} is the one-step-ahead prediction error of the observations of regime k at time t, F_{kt} its variance and d_{kt} the number of locations of the regime observed at time t (Section 3.1 treats the gaps). The log-likelihood is maximized by the EM algorithm[2],

which treats the latent states as missing data and is accelerated by SQUAREM[37]. The E-step runs the Kalman smoother. The M-step updates m_{0k}, G_k, $\Sigma_{\eta k}$, $\sigma^2_{\omega k}$ and β_k in closed form, and the noise-to-signal ratio $\sigma^2_{\epsilon k}/\sigma^2_{\omega k}$ and the range parameter θ_k by a Newton–Raphson step on the logarithmic scale[5,7]. See Section A1 of Computational and Algorithmic Supplement A for the steps and the stopping rule.

2.2 | Spatial cohesion of the regimes

The partition is estimated together with the parameters, as in clusterwise regression[22,23]. Left alone, such a partition need not respect geography: two distant locations may join the same regime only because their series happen to fit it. Following Sugasawa and Murakami[24], we add a Potts-type penalty[25] that rewards neighboring locations for sharing a regime, and estimate the labels and the parameters by maximizing

$$Q\big(k_1, \dots, k_d; \Psi_1, \dots, \Psi_K\big) \;=\; \sum_{i=1}^{d} \ell_i(k_i) \;+\; \phi\, c \sum_{(i,j)\in E} I(k_i = k_j), \tag{7}$$

where $\ell_i(k)$ is the score of location i under regime k (Section 2.3), E is the edge set of a graph of neighbors of the locations, described below, $\phi \geq 0$ is the penalty, and $c > 0$ is a scale factor. Each location contributes a whole series to the likelihood, so the differences between its scores grow with T and with the units of the response. The scale

$$c \;=\; \frac{\operatorname{median}_i \Delta_i}{\bar{m}}, \qquad \Delta_i = \max_k \ell_i(k) - \min_k \ell_i(k), \tag{8}$$

with $\bar{m}$ the mean number of neighbors in the graph, is computed once, at the first sweep, on the regimes of the unpenalized fit. It lets one grid of penalties serve data sets of any length and scale: at $\phi = 1$, full agreement with the neighborhood is worth about as much as the typical gain from the best-fitting regime (see Section A3 of Computational and Algorithmic Supplement A).

Notice that the penalty encourages spatial cohesion without imposing it: a location that fits a distant regime much better can still leave its neighborhood. At $\phi = 0$ the estimation is plain clusterwise estimation, and as ϕ grows the partition tends to a few contiguous blocks. Moreover, the Potts term originally comes from lattices[24,35] and areal units[34,36]: on point-referenced data it needs only a graph of neighbors, and we use the m-nearest-neighbor graph of the locations, built with the same distance as the covariance in Equation (4), geodesic for geographic coordinates. The graph is symmetrized, so that the reward of an edge does not depend on which of its two endpoints is being updated, and E counts each pair once. In a network of irregular density, a nearest-neighbor graph keeps the number of neighbors roughly constant, so that the penalty weighs about the same for every location. Two features limit the role of the penalty. First, ϕ is a tuning constant, chosen by comparing the likelihoods that the partitions estimated at different penalties reach (Section 3.2), and the data can decline it by selecting $\phi = 0$. Second, the existence of regimes is decided by comparing the likelihood of the partition with that of the pooled model, a comparison in which the penalty does not enter. A spatially compact partition is not, by itself, evidence of regimes. See Section A8 of Computational and

Algorithmic Supplement A for a discussion of the choice of the graph, of the double role of spatial proximity in the covariance and in the penalty, and of the normalizing constant of the Potts term.

2.3 | The assignment score: a conditional pseudo-likelihood

The objective function in Equation (7) needs a contribution $\ell_i(k)$ of each location i under each candidate regime k. In the cross-sectional models from which the apparatus comes, this contribution is exact. In clusterwise regression[24] and in the spatially clustered Fay–Herriot model[36], the units are independent given the parameters of their cluster, so the likelihood factorizes, $\ell_i(k)$ is the log-density of unit i under cluster k, and Q is a penalized likelihood. In SC-STEM no such contribution exists. The measurement covariance in Equation (4) couples the locations of a regime, so the joint density of z_t does not factorize across them. Moreover, the candidate regimes have different members, hence different covariance matrices, so even a partial factorization would not be comparable across regimes.

We therefore use a conditional pseudo-likelihood. Let $\hat{y}^{(k)}_{1:T}$ be the smoothed latent states of regime k at its current parameters. The score of location i under regime k is

$$\ell_i(k) \;=\; -\frac{T}{2}\,\log\!\left(2\pi v_k\right) \;-\; \frac{1}{2v_k}\sum_{t=1}^{T}\left(z_{it} - x_{it}^{\top}\beta_k - A_i\hat{y}_t^{(k)}\right)^2, \qquad v_k = \sigma^2_{\epsilon k} + \sigma^2_{\omega k}. \tag{9}$$

Equation (9) represents the marginal log-density of the series of location i given the latent path of regime k: the conditional mean is that of the measurement equation, Equation (1), at the smoothed state, and v_k is the diagonal element of $\Sigma_{e,k}$, the total residual variance of one location. What the score leaves out is the covariance between location i and its neighbors within the regime, and the uncertainty of the smoothed state; it is exact when the spatial correlation is negligible. The construction is a composite likelihood of the independence type[38]: like the pseudo-likelihood of Besag[39], it replaces an intractable joint density with a product of low-dimensional ones.

The score is a good ranking device. It is comparable across regimes, since it is evaluated on the same data with the same form. It responds to every component of Ψ_k that matters for location i: β_k through the mean, G_k and $\Sigma_{\eta k}$ through the smoothed state, and the variance components through v_k. It costs a single pass over the series. It is not, however, the exact likelihood that the parameter step maximizes.

2.4 | Estimation strategy: alternating optimization

The objective function in Equation (7) is maximized by alternating between its two blocks of arguments: given the labels, the parameters of each regime are estimated by running the EM algorithm of Section 2.1 on the sub-network of that regime; given the parameters, the labels are updated one at a time. The scheme is summarized in Algorithm 1.

Algorithm 1 Estimation of the SC-STEM model at fixed K and ϕ

Input: data $\{z_{it}, x_{it}\}$, coordinates $s_1, \dots, s_d$, loading matrix A, number of regimes K, penalty ϕ, number of neighbors m, minimum cluster size $n_{\min}$.
Build the symmetrized m-nearest-neighbor graph and its edge set E.
if $\phi = 0$: fit the pooled STEM model ($K = 1$) and initialize the partition by k-means on the departures of each location from it (see Section A3 of Computational and Algorithmic Supplement A), compressed by principal components, with multiple restarts and a minimum-size filter; **if** $\phi > 0$: initialize the partition at the one estimated by this algorithm with $\phi = 0$ and the same K.
repeat
    Parameter step. For $k = 1, \dots, K$, estimate Ψ_k by EM on the locations currently assigned to regime k; a regime below $n_{\min}$ retains its last valid parameters.
    Compute the score matrix $\{\ell_i(k)\}$ of Equation (9) and, at the first sweep only, the penalty scale c in Equation (8).
    Label step (ICM). Visit the locations in turn; set $k_i \leftarrow \arg\max_k \{\ell_i(k) + \phi c \sum_{j:(i,j)\in E} I(k = k_j)\}$, subject to the minimum-size constraint, using the labels of the other locations as currently updated.
    Swap pass. Exchange the labels of pairs of locations in different regimes whenever the exchange strictly increases Q.
    Evaluate Q and record the configuration if it improves on the best one visited so far.
until the partition is unchanged, the improvement of Q falls below tolerance, a previously visited partition reappears, or the iteration budget is exhausted.
Final refit. Re-estimate the K regimes once on the best partition visited, and report all estimates and information criteria from this refit.

The label step is the sequential update of the iterated conditional modes (ICM) algorithm of Besag[40]: the locations are visited in turn and each label maximizes its own penalized contribution given the current labels of all the others, so that within a sweep, and at fixed parameters, the objective cannot decrease and label cycling is ruled out by construction. The alternative used in the cross-sectional literature[24], in which all labels are refreshed simultaneously given the previous configuration, is faster per sweep but offers no such guarantee.

3 | COMPUTATIONAL, ALGORITHMIC AND INFERENTIAL ASPECTS OF SC-STEM

This section completes Algorithm 1 by describing the most relevant aspects that matter for practical and interpretative purposes, namely the assignment score, with its consequences for the size of the regimes and the convergence of the algorithm, the final refit and the cost of a fit, and the treatment of missing observations (Section 3.1), the selection of the number of regimes and of the penalty (Section 3.2), and inference through a bootstrap procedure that accounts for the estimated partition (Section 3.3). Computational and Algorithmic Supplement A gives the remaining computational and algorithmic details: the EM algorithm for STEM and within SC-STEM, with the two accelerations available for it, that is, the ECME algorithm[41] and SQUAREM[37], and their stopping rule (see Section A1); the derivation of the treatment of missing observations (see Section A2); the initialization of a fit and the data-driven scale of the penalty (see Section A3); the control of the regime sizes, which keeps the partition from collapsing (see Section A4); the behavior of the information criteria (see Section A6); and inference on several coefficients under the bootstrap (see Section A7).

3.1 | The assignment score, the final refit and missing data

The assignment score of SC-STEM (see Section 2.3) raises two relevant issues, namely the size of the regimes and the convergence of the alternation. As for the size, we recall that an estimated partition can degenerate, much as the likelihood of a Gaussian mixture can be made unbounded by a component that concentrates on a few observations [19]. The score in Equation (9) rewards a small residual variance: as locations move to the regime that fits them best, the most homogeneous regime becomes more homogeneous still and, left alone, would absorb the whole network, even at $\phi = 0$. Each regime must also contain enough locations to support its own fit. Algorithm 1 therefore bounds the size of a regime from below by $n_{\min}$, which, given d, also bounds the number of regimes that can be entertained, and adds a swap pass that exchanges labels without changing the sizes, so that the constraint does not freeze the partition (see Section A4 of Computational and Algorithmic Supplement A).

The second issue is the convergence of the alternation between the two steps, which is not monotone. Each step is monotone in its own right, but the trace of Q may decrease from one iteration to the next: the label step maximizes the penalized *pseudo*-likelihood in Equation (7), the parameter step the *exact* likelihood of each regime, and the two would coincide only if the likelihood factorized over the locations, which the spatial covariance in Equation (4) prevents. Non-monotonicity is thus a property of the spatio-temporal specification, without counterpart in the cross-sectional spatially clustered models. The algorithm returns the best configuration visited rather than the last, and refits on it (see Section A5 of Computational and Algorithmic Supplement A for the argument, the stopping rule and the diagnostics).

When the alternation stops, the regimes are re-estimated once on the best partition visited, holding the labels fixed. Each regime starts from its estimates of the last iteration of the alternation, and the EM algorithm runs with the tolerances of the pooled fit, so that the log-likelihoods compared across K are computed to the same accuracy (see Section A1.5 of Computational and Algorithmic Supplement A). The refit depends on the data, the partition and these settings, not on ϕ, so the fits of a grid that end at the same partition share one refit. All reported coefficients, variance components and information criteria come from this refit and are based on the *exact* clusterwise log-likelihoods returned by the EM algorithm, not on the pseudo-likelihood in Equation (9) used to rank regimes during the assignment step. The number of free parameters of a configuration is

$$\mathrm{df} \;=\; K_{\mathrm{eff}}\,(r + 3 + 3p)\,, \tag{10}$$

for a diagonal specification of G and Σ_η with r covariates and latent dimension p. Here K_{eff} counts the regimes that could actually be re-estimated, each with r regression coefficients, the three parameters $(\sigma^2_\epsilon, \sigma^2_\omega, \theta)$ of the measurement covariance, and p parameters each for G, Σ_η and m_0.

This refit, like every parameter step, runs K independent EM algorithms, one per regime, and these determine the cost of a fit. Within a regime of size d_k, the dominant cost is the factorization of the $d_k \times d_k$ measurement covariance at each EM iteration, that is $O(d_k^3)$, so that partitioning the network is computationally advantageous relative to the pooled fit whenever the regimes

are of comparable size (i.e., $\sum_k d_k^3 \ll d^3$ for any non-trivial partition). The label step is a single sweep over the d locations with cost $O(d\ K\ \bar{m})$, where $\bar{m}$ is the average number of neighbors, and the score matrix in Equation (9) costs $O(d\ K\ T)$.

Finally, gaps in the response are handled by the state-space form of each regime, following Durbin and Koopman[4] Sections 2.7 and 4.10: at each time point the Kalman filter uses the observed sub-vector of z_t, a time point with nothing observed contributes no term, the likelihood remains the exact likelihood of the observed values, and the smoother needs no change. Because $\Sigma_{e,k}$ couples the locations, the conditional expectation of a missing value is not its signal alone, as it would be with a diagonal measurement covariance, but its signal plus the part of the measurement error predicted from the locations observed at the same time point: denoting by O and M the observed and the missing locations, $P_t r_{O,t}$, where $P_t = [\Sigma_{e,k}]_{MO}([\Sigma_{e,k}]_{OO})^{-1}$ and $r_{O,t}$ is the residual of the observed locations at the smoothed state. Ignoring that term would bias the variance components downward. The assignment score is evaluated on the T_i periods at which location i was observed, the bootstrap of Section 3.3 reimposes the observed pattern of gaps on every replicate, and every location must retain at least one observation (see Section A2 of Computational and Algorithmic Supplement A). We stress that gaps are allowed in the response only: the covariates and the coordinates enter the closed-form M-step and the distance matrix directly, so their gaps would require a stochastic E-step; they must be imputed before the model is fitted, as in the application (see Section 5.1).

3.2 | Tuning and model selection

The number of regimes K and the penalty ϕ are hyperparameters governing the complexity of the model, and they are selected jointly over a grid by the Bayesian information criterion (BIC). For each configuration the exact total log-likelihood ℓ of the final refit is recorded, together with the parameter count in Equation (10), and

$$\mathrm{BIC} = -2\ell + \log(N)\,\mathrm{df}, \tag{11}$$

where N is the number of observations entering the likelihood ($N = dT$ when the panel is complete).

Unlike the Akaike information criterion (AIC), the BIC is consistent as the series lengthen: the likelihood gain that a superfluous regime extracts from the search over partitions does not grow with T, while the BIC charges each additional regime $\log(dT)$ times its number of parameters. The BIC is therefore consistent as the series lengthen on a fixed network, and it overselects only when the network is wide relative to $\log(dT)$. Other criteria can take the place of the BIC in the rule below (see Section A6 of Computational and Algorithmic Supplement A for how they behave as T and d grow, and Section B6.7 of Simulation Supplement B for a comparison of seven of them in the simulation study).

Two caveats apply to any likelihood-based comparison in this setting. First, the partition is itself optimized on the data, so the maximized likelihood retains an optimism bias that a parameter count of the form in Equation (10) does not correct: the degrees of freedom spent on the partition are not counted at all. Second, by the mechanism described in Section 2.4, the

clusterwise variance components shrink as the assignment step pursues homogeneity, which inflates the likelihood of exactly those configurations in which the partition is least constrained.

The selection rule for K and ϕ.

The selection rule chooses (K, ϕ) in two steps over the grid. In the first, for each penalty in a moderate band Φ_M, it records the BIC-minimizing K among the admissible configurations at that penalty and the pooled model, whose criterion does not depend on ϕ; $\hat{K}$ is the modal winner across the band, ties resolved toward the smaller K, as in Maranzano et al.[36]. Excluding $\phi \approx 0$ removes the configurations in which the optimism bias of the caveats above is largest, and since the pooled model competes at every penalty, a partition is adopted only when it improves on it. In the second step, at $\hat{K}$, the rule takes the admissible fit with the smallest BIC over the whole grid, ties resolved toward the smaller ϕ, as in Cerqueti et al.[34]. At a given K the parameter count does not depend on ϕ, so the step compares the likelihoods that the partitions estimated under different penalties reach. Every penalized fit starts from the unpenalized one (see Section A3 of Computational and Algorithmic Supplement A): a penalty is selected only when it has moved the partition to one of higher likelihood, and the optimism bias, which favors $\phi = 0$, makes the step conservative. The criterion is built on the likelihood alone, so the normalizing constant of the Potts term never enters it. The default band is $[0.025, 0.2]$; the strong values 0.5 and 1 of the default grid lie outside it and compete in the second step only, where the likelihood decides whether they are used. Finally, only *admissible* configurations enter any comparison, that is, those whose final partition has no empty or undersized regime (see Section A4 of Computational and Algorithmic Supplement A). This is not a formality here: with d locations and the default minimum regime size of $r + 2$, the number of regimes that the network can support at all is bounded by $d/(r+2)$, and configurations pressed against that bound have partitions determined as much by feasibility as by fit.

3.3 | Bootstrap inference

Conditionally on the partition, an SC-STEM fit reduces to a collection of ordinary STEM fits, for which the parametric bootstrap of Fassò and Cameletti[7] applies unchanged. Conditioning on the partition, however, treats as known something that was estimated, and understates the uncertainty of every quantity that depends on it. We therefore adopt a *refit-with-clustering* parametric bootstrap strategy, which re-runs the entire procedure (including the endogenous partitioning) at every draw. Repeating the clustering on every resample is the idea behind the bootstrap assessment of cluster stability[42] and the parametric bootstrap of finite mixture regression models[43]; Maranzano et al.[36] used it for spatially clustered small area models. See Algorithm 2 in Appendix A for the procedure.

Given the fit at the selected $(\hat{K}, \hat{\phi})$, each bootstrap sample is generated *regime by regime*: for each regime k, a spatio-temporal data set is simulated from the STEM model of that regime evaluated at $\hat{\Psi}_k$, using the observed covariates, coordinates and

loading rows of the locations assigned to it, so that both the within-regime spatial covariance and the latent temporal dynamics of the fitted model are reproduced. The original ordering of the locations is then restored, so that the simulated data are aligned with the rows of the neighborhood graph and every location keeps its own neighbors; the spatial structure of the design is thus preserved, with the fixed-effects surface inheriting the contiguity of the estimated regimes. The full procedure is finally re-run on the bootstrap sample at the same $(\hat{K}, \hat{\phi})$ and with the same algorithmic settings. See Section A7 of Computational and Algorithmic Supplement A for how the regimes of every draw are aligned with the estimated ones, for the diagnostics of the draws, and for the intervals and tests computed from them.

4 | SIMULATION EXPERIMENTS

We use simulations to assess whether SC-STEM does what it is designed for, that is, to find latent spatial regimes when they exist, to describe them correctly, and not to identify them when they do not exist. We examine three outputs. The number of regimes comes first, since every other estimate is conditional on it. The partition and the regime-specific parameters come next and show which locations belong together and how the regimes differ. Last, we examine how both react to departures from the assumptions of the model. Every replication runs the whole procedure, from the grid of (K, ϕ) to the selected fit.

4.1 | Design

The data are generated by a single latent process loading equally on every location, one covariate and $K_0 = 3$ true regimes. Within regime k, the data are simulated as

$$z_{it} = \beta_{0k} + \beta_{1k}\, x_{it} + y_t^{(k)} + e_{it},$$

with $y_t^{(k)}$ a first-order autoregression with persistence G_k and innovation variance $\sigma^2_{\eta k}$, and e_{it} the sum of a nugget $\sigma^2_{\epsilon k}$ and of an exponential spatial field with partial sill $\sigma^2_{\omega k}$ and range parameter θ_k.

Regime 1 (baseline) is the same in every scenario, and its response has variance about one, so that every contrast between regimes reads in standard deviations of the response. Three groups of scenarios are run:

S2, no regimes: the pooled STEM model at the baseline;

S1, complete heterogeneity: every block of parameters differs across the regimes, strongly (S1s) or weakly (S1w): from regime 1 to regime 3 the complexity rises: the covariate matters more, shocks die out faster but the latent signal is larger and the error variance grows, by much in S1s and by little in S1w, while the range shortens from half the network to a quarter of it at both levels (see Section B3 of Simulation Supplement B);

S3, one common block: the regression coefficients (S3beta), the persistence (S3G), the innovation variance (S3Seta), the range (S3theta) or the whole measurement error (S3error) is common to the regimes, the other blocks as in S1s.

While the groups differ in which parameters set the regimes apart, we also consider two variants that differ in how the regimes are linked to one another. In the variant `ind` the regimes have independent latent processes and spatial fields, as SC-STEM assumes. In the variant `shr` they share the latent shocks, which each regime filters with its own parameters, and, where the range is common, one spatial field, so that the regimes separate through their parameters only. The variant `shr` is thus a realistic departure from the model: the stations of different regimes, for instance, face the same cost shocks. The scenario-variants also differ in how clearly the data separate the regimes, that is, in how much evidence a single location carries about the regime it belongs to. This separation is induced by the design, through the parameter values of the scenarios and the dependence of the variants, which were fixed to favor interpretability. We quantify the degree of separation by the expected log-likelihood ratio that a location accumulates in favor of its own regime against the nearest wrong one (see Section B4 of Simulation Supplement B for its derivation and values). By this measure, S1w-`shr`, where every contrast is small, is the hardest case, as intended, and S3beta-`shr`, where the regimes share their regression coefficients, is almost as hard.

Overall, the nine scenario-variants are S2; S1s in both variants, S1s-`ind` and S1s-`shr`; S1w-`shr`; and the five variants S3beta-`shr`, S3G-`shr`, S3Seta-`shr`, S3theta-`shr` and S3error-`shr`.

The scenario-variants are crossed with the number of locations, $d \in \{40, 100, 200, 400\}$, the length of the series, $T \in \{60, 120, 365\}$, four geometries of the regimes in space, from total overlap to strong separation, balanced or unbalanced regime sizes (in the unbalanced case the three regimes hold one sixth, one third and one half of the locations), and the number of neighbors of the Potts graph, $m \in \{3, 5, 10\}$. The margins form five blocks around the reference cell (i.e., $d = 100$, $T = 120$, regimes interpenetrating along their borders, balanced sizes, $m = 5$), for 332 cells and 200 replications per cell. The number of regimes $\hat{K}$ is the one selected by the two-step rule, to be compared with the true number $K_0 = 3$. The partition and the parameters are read off the fit with K fixed at the true value K_0 (and $\phi = 0.05$), whatever $\hat{K}$ is, and the estimated partition is compared with the true one, that is, with the regimes that generated the data. In this way a poor estimate is not confounded with a poor selection. See Appendix B for examples of simulated data, and Simulation Supplement B for the full design and the extended results.

4.2 | Results

Figure 1 condenses the main results of the simulations, that is, the selection of K over the whole core of the design, and the partition and the parameters at the reference cell. See Simulation Supplement B for the results cell by cell.

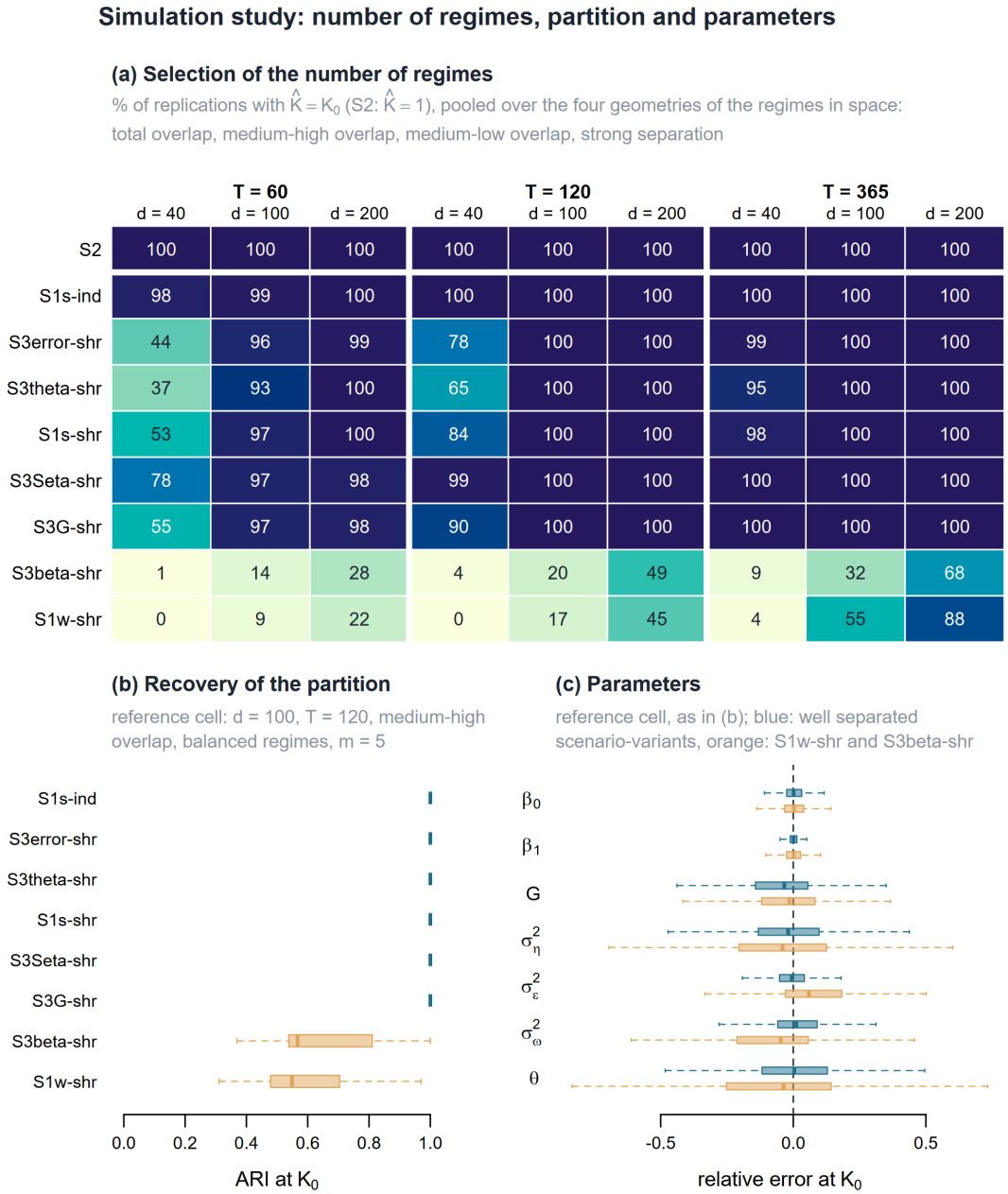


FIGURE 1 Simulation results. Panel (a): Percentage of the replications in which the two-step rule selects the true number of regimes, $\hat{K} = K_0$ ($\hat{K} = 1$ in S2), over the four geometries (4×200 replications per entry; S2, 200), by T and d; the scenario-variants run from the most to the least separated. Panel (b): Adjusted Rand index between the true partition and the partition estimated by the fit with K fixed at the true K_0, at the reference cell (i.e., $d = 100$, $T = 120$, medium-high overlap, balanced sizes, $m = 5$). Panel (c): Boxplots of the *relative error* (i.e., (estimate − truth)/|truth|), of the parameters at K_0, the three regimes together, at the reference cell, for the seven well-separated scenario-variants (blue) and for S1w-`shr` and S3beta-`shr` (orange).

Number of regimes.

- *Homogeneous networks are rarely or never split.* When the data contain no regimes (S2), the two-step rule keeps the pooled model in nearly every replication, on small and large networks alike. Spurious regimes appear only as the setting becomes

more complex: when weak regimes overlap in space on large networks, the rule sometimes adds one regime to the true ones, the risk anticipated in Section 3.2.

- *Well-separated regimes are found unless the sample is very small.* Their number is recovered almost always from one hundred locations on, and also on the smallest networks once the series are long. The failures occur with few locations and short series, and the rule merges regimes rather than inventing them (under-selections).
- *Weakly separated regimes need data.* When the regimes differ little (S1w-`shr`) or share their regression coefficients (S3beta-`shr`), the rule often merges two of them; the share of correct selections grows steadily with the number of locations and with the length of the series.

Partition and parameters.

- *With the right number of regimes, the estimates are accurate.* The partition of well-separated regimes is recovered almost exactly, and that of weakly separated regimes improves with d and T. The estimates of every regime are centered on the true values (Figure 1 c). The regression coefficients are the most precise; the persistence, the innovation variance and the range, which only the dynamics and the spatial structure of the data identify, are the least precise.
- *A wrong number of regimes affects the variance components more than the coefficients.* When the rule selects the wrong number of regimes, the coefficients of the fit at K_0 stay centered but are more dispersed, while the variance components are distorted (see Section B6.4 of Simulation Supplement B).
- *The selected model recovers the signal better than the pooled model* (Figure 2). Whatever number of regimes is selected, the fitted signal is closer to the true conditional mean than that of the pooled STEM model. When the rule merges two weak regimes, the merged model fits the signal better than the fit with the true number of regimes: merging is a sensible response to data that cannot tell the regimes apart.

Effect of the departures.

- *The regression coefficients drive the separation.* Independent regimes (S1s-`ind`) are identified in practically every case. When the regimes share their shocks, they must be told apart by their parameters, and the coefficients are the main lever: removing their contrast (S3beta) costs almost as much as weakening every contrast (S1w), while a common persistence, innovation variance, range or error leaves the procedure close to S1s-`shr`.
- *Regimes that are compact in space are harder to count than overlapping ones.* A plausible reason is that the locations of a compact regime share one spatial field, so they carry less independent information, and the field can mimic a contrast between regimes. The estimated partition, instead, hardly depends on the geometry. Overlap acts the other way on over-selection, which occurs with overlapping weak regimes only (see Sections B6.2 and B6.3 of Simulation Supplement B).

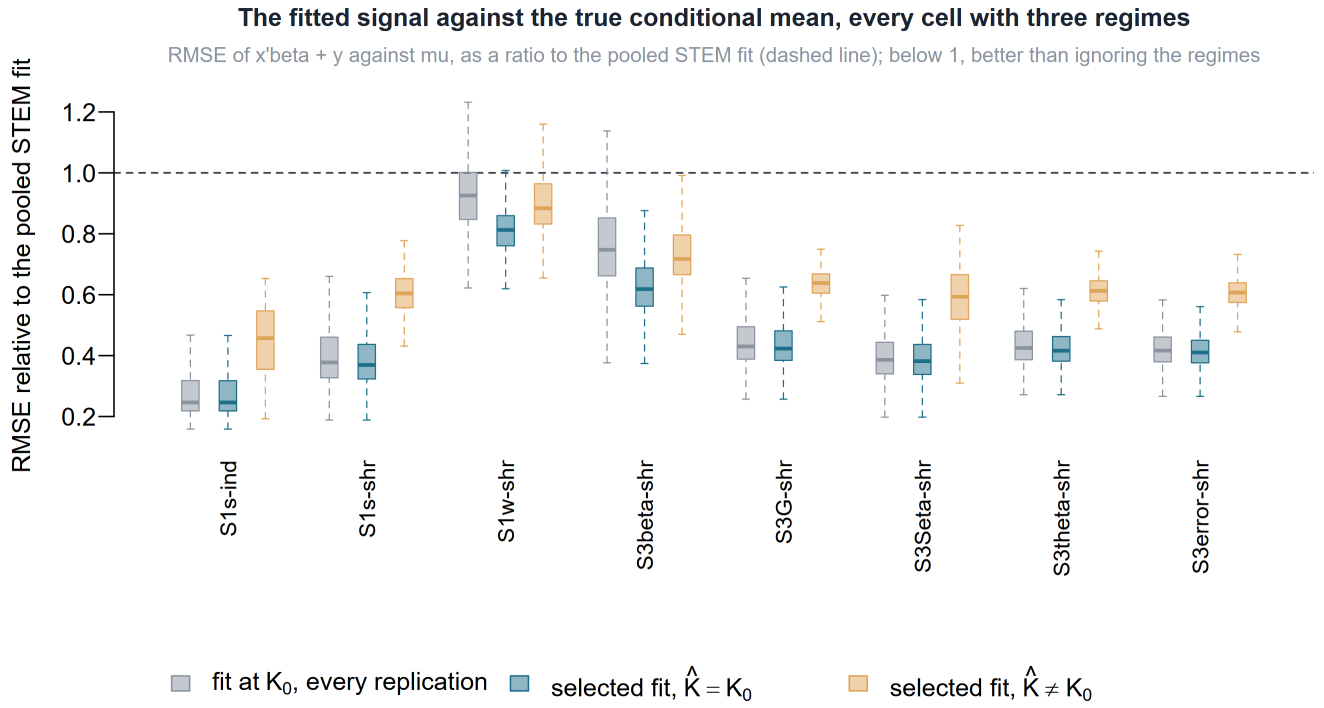


FIGURE 2 Signal extraction: root mean squared error of the fitted signal against the true conditional mean, as a ratio to that of the pooled STEM fit (dashed line), over every cell with three regimes. The fitted signal is computed as $\hat{\mu}_{it} = \hat{\beta}_{0g} + \hat{\beta}_{1g} x_{it} + \hat{y}_t^{(g)}$, with g the estimated regime of location i and $\hat{y}_t^{(g)}$ the smoothed latent path, whereas the true conditional mean is $\mu_{it} = \beta_{0k} + \beta_{1k} x_{it} + y_t^{(k)}$ of location i in its true regime k, that is, the expectation of z_{it} given the covariate and the latent path. Gray: the fit at K_0, in every replication; blue and orange: the selected fit, in the replications with $\hat{K} = K_0$ and $\hat{K} \neq K_0$. Below one, SC-STEM recovers the signal better than the pooled model.

- *Unbalanced sizes affect the weak regimes only, and the number of neighbors of the Potts graph changes nothing.* Moderate penalties leave the partition unchanged, larger ones oversmooth it (see Sections B6.5 and B6.6 of Simulation Supplement B).

5 | APPLICATION: LOCAL PRICE LEADERSHIP IN ITALIAN URBAN FUEL MARKETS

The retail fuel industry is a standard setting for the study of local competition and of cost pass-through[44]: stations sell a nearly homogeneous product, compete mostly with a handful of nearby competitors[45,46], and post prices that are observed station by station. A question that recurs in this literature is whether a station's price follows that of its neighbor, and whether the stations of one brand lead the others[47]. It is a question about heterogeneity as much as about dependence. A model with one relation per city answers it on average, and an average over groups of stations that behave differently can be close to zero, or significant, for reasons that hold in some groups only; a test run station by station, at the other extreme, might deal with a few observations without borrowing of strength. Between these two extremes lies grouped heterogeneity[26,27], which SC-STEM provides for fuel

stations: the stations of a city may relate to their neighbors in different ways, but only across a few regimes, found from the prices, whose members share one relation and pool their information to estimate it.

In this spirit, we address three research questions, separately in each of the eleven cities of our data.

Q1, local leadership. Do the past relative prices (Equation (12) in Section 5.1) of the nearest competitor X help to predict those of a pump Y, given the past of Y, the past of its other fuel, the changes of the excise duties and the shocks common to its regime? This is Granger non-causality[48] of X for Y. We test it regime by regime with a bootstrap Wald test on the coefficients of the lags of X (see the note to Table 1 for analytical details). The size of the leadership is measured by the long-run pass-through from X to Y, with its bootstrap interval.

Q2, heterogeneity. Is the answer the same for all the pumps of a city, as a pooled model would assume? If not, what are the groups that answer differently: submarkets in space, or pricing strategies of the brands? The number of regimes $\hat{K}$, the penalty $\hat{\phi}$ and the composition of the regimes answer it, and the pooled STEM model, tested in the same way, gives the answer a model without regimes would give.

Q3, fiscal pass-through. Do the pumps of a city pass the changes of the excise duties through at the same speed? The coefficients of the fiscal pulses, specific to each regime, answer it.

5.1 | Data and settings

We consider weekly prices of gasoline and diesel at the fuel station level, published by the Italian Ministry of Enterprises and Made in Italy on the Osservaprezzi Carburanti platform[49], running from January 3, 2022, to June 28, 2026 (i.e., 234 weeks overall), each represented by its Sunday price, and 2021 serves as a presample for the lags. Comprehensive information on the daily data provided by the platform is described by Demir[50]. The stations are pre-treated so as to keep those that set their prices actively and report them regularly (i.e., codes of the same site linked, highways excluded, at least 95% of the days reported, no price left unchanged for more than 90 days), which leaves 12,246 stations in Italy. Among those, we consider a subset of stations located throughout the major Italian cities and with known brand and georeferenced coordinates. The final data set covers 3,136 fuel stations (of which 264 are independent or *pompe bianche*) operating across eleven metropolitan cities, namely Rome (653 stations, maximum), Turin, Milan, Venice, Bologna, Florence, Naples, Bari, Palermo, Catania and Messina (118 stations, minimum).

The response of a pump is its *relative price*, in cents per liter, that is, the Sunday price net of the national mean of the week, which removes the common stochastic trend of costs and taxes, and of the mean of the pump, which removes its permanent premium. Formally, let p_{it} be the Sunday price of a fuel at pump i in week t, $\bar{p}_t$ the mean price of the fuel over all the non-highway stations of Italy that report it in week t (some 20,000), and $\mathcal{T}_i$ the set of the T_i weeks of the window in which pump i

reports a price. The relative price is

$$z_{it} = (p_{it} - \bar{p}_t) - \frac{1}{T_i} \sum_{s \in \mathcal{T}_i} (p_{is} - \bar{p}_s). \tag{12}$$

See Appendix C for the two steps applied to two stations of Rome (Figure C1). After the double centering, the series fluctuate around zero and become stationary (relative prices revert to their mean, and the augmented Dickey–Fuller test rejects a unit root for 96 to 100% of the pumps of every city). Notice that the first step matters in a window that contains large shocks common to all stations, such as the war in Ukraine[51] and the closure of the Strait of Hormuz[52], in 2022 and 2026.

The neighbor X of a pump Y is its nearest station of a given group, which defines three settings:

(1) Y a station of a major brand (Eni, Api-Ip, Esso, Q8, Tamoil), X the nearest independent station;

(2) Y an independent station, X the nearest station of a major brand;

(3) Y a station of a major brand, X the nearest station of another major brand.

Settings 1 and 3 have 2,427 pumps Y in all, setting 2 has 264, ranging from 6 in Florence to 71 in Naples. Setting 3 is the dense market, the nearest station of another major brand lying at a median of 0.6 to 1.6 kilometers; independents are rare. In setting 1 the nearest one lies at a median of 1.4 kilometers in Naples and of 2.8 to 7.9 kilometers elsewhere. Settings 1 and 2 read the same pairs in the two directions. Every setting, city and fuel is a separate model (see Section C1 of Application Supplement C for details).

The missing weeks of the series that enter the lags are completed beforehand by a pooled STEM model of the city with the intercept and the pulses (Section 3.1), since the regimes are not known at that point. The response keeps its missing weeks, which the Kalman filter handles.

5.2 | Empirical strategy

The empirical specification of the SC-STEM models is determined based on some exploratory insights. The partial autocorrelation function of the relative price, computed pump by pump, exceeds the band $\pm 2/\sqrt{T}$ for all the pumps at lag 1, for 56–72% at lag 2, 26–41% at lag 3 and 14–22% at lag 4 (medians over the cities), and for at most 17% from lag 5 on. The order of the autoregression chosen by the AIC pump by pump has median 3 to 4. The partial autocorrelation at lag 52 (i.e., yearly tendency) is within the band for 99–100% of the pumps, so the autocorrelation at that lag reflects the persistence of the series, not an annual season. The own dynamics therefore reach lag 4, and we use four lags for every block of regressors.

For a pump i of a city and week t, let z_{it} be the relative price of the fuel (either gasoline or diesel), o_{it} the relative price of the other fuel at the same pump, and x_{it}, w_{it} the relative prices of the two fuels at the neighboring pump $j(i)$ of the setting. For i in

regime k, the observation equation is

$$z_{it} = \beta_{0k} + \sum_{l=1}^{4} \left(b_{kl} z_{i,t-l} + c_{kl} o_{i,t-l} + d_{kl} x_{i,t-l} + e_{kl} w_{i,t-l} \right) + \gamma_k' \pi_t + y_t^{(k)} + \epsilon_{it}. \tag{13}$$

In Equation (13), π_t collects the pulses of the changes of the excise duties (the week of each change and the next), $y_t^{(k)}$ is the latent AR(1) process of the regime, which absorbs the shocks common to its pumps, and ϵ_{it} is the measurement error of the regime, the sum of a nugget and of a spatially correlated component with exponential correlation, as in Equation (4).

The neighbor leads pump i in regime k when the eight coefficients $(d_{kl}, e_{kl})_{l=1}^{4}$ are not all zero. The test of Granger causality is thus run within groups of stations, as it has been between spatial clusters of events in space and time[53]; here, however, the groups are the regimes, estimated jointly with the relation that each of them shares. Notice that the test says whether the neighbor leads, not by how much. The lag coefficients are hard to read one by one: they spread the response of the pump over four weeks, and its own lags carry the response further. The long-run pass-through condenses them into a single number with an economic meaning. If the relative prices of the pump and of its neighbor settled at constant levels z^* and x^*, with the other regressors held fixed, Equation (13) would give $z^* = \text{const} + z^* \sum_l b_{kl} + x^* \sum_l d_{kl}$, hence

$$\lambda_k = \frac{\partial z^*}{\partial x^*} = \frac{\sum_{l=1}^{4} d_{kl}}{1 - \sum_{l=1}^{4} b_{kl}}, \tag{14}$$

defined when $\sum_l b_{kl} < 1$, as for a mean-reverting relative price. The long-run pass-through quantifies the change (in cents) of the relative price of a pump of regime k when that of its neighbor on the same fuel moves permanently by one cent. A value of one would mean that the pump fully aligns with its neighbor, a value of zero that it ignores it. It thus measures the strength of local leadership, which is what matters for competition: how much of a price change of one station reaches its neighbors.

Each model is estimated in two stages.

(i) *The regimes.* For a given city, the grid of (K, ϕ), with $K \leq \min(5, \lfloor n/5 \rfloor)$ for n pumps, so that every regime has at least five, and ϕ on the default grid, and the two-step rule of Section 3.2 on the BIC give $(\hat{K}, \hat{\phi})$ and the partition $\hat{\mathcal{G}}$. Below ten pumps the model is the pooled STEM model.

(ii) *Inference.* The refit-with-clustering bootstrap of the fit at $(\hat{K}, \hat{\phi})$, with $B = 100$ draws, gives the Wald statistic W_k of every regime and the percentile interval of λ_k. Bootstrap tests are common in the analysis of Granger causality[54,55], and here the bootstrap also carries the uncertainty of the partition into the test. The pooled STEM model is fitted, bootstrapped and tested in the same way. As a benchmark, the Granger test is also run pump by pump by least squares, in its classical F form and with the heteroskedasticity-consistent covariance HC1[56,57], and combined over the pumps of a city by the test of Dumitrescu and Hurlin[58].

TABLE 1 Synthesis of the 66 models by setting and fuel. *Regimes*: total number of regimes over the eleven cities. $\hat{K}$: range of the selected number of regimes across the cities. *X leads Y*: share of the regimes, and of the pumps Y in them, where the bootstrap Wald test rejects $d_{k1} = \cdots = e_{k4} = 0$ at 5%: the test statistic is $W_k = \hat{\vartheta}_k^\top \hat{V}_k^{-1} \hat{\vartheta}_k$, with $\hat{\vartheta}_k = (\hat{d}_{k1}, \ldots, \hat{d}_{k4}, \hat{e}_{k1}, \ldots, \hat{e}_{k4})^\top$ being the estimated coefficients of the lags of X in Equation (13) and $\hat{V}_k$ being the covariance of their aligned bootstrap draws (Section 3.3). The statistic is referred to the χ^2_8 distribution. *Pooled model*: cities in which the same test on the pooled STEM model rejects. *Estimated long-run pass-through*: median and interquartile range over the regimes, and share of the regimes whose bootstrap 95% interval lies above zero. *Per pump*: share of the pumps for which the least-squares HC1 Granger test rejects, median over the cities. Setting 1: Y major brand, X independent; 2: Y independent, X major brand; 3: Y major brand, X another major brand.

					X leads Y (%)		pooled	long-run pass-through		per pump
setting	fuel	pumps	regimes	$\hat{K}$	regimes	pumps	model	median (IQR)	> 0 (%)	HC1 (%)
1	gasoline	2427	53	4–5	62	67	7/11	0.08 (0.02, 0.14)	32	31
	diesel	2427	52	4–5	73	77	8/11	0.05 (0.00, 0.09)	31	34
2	gasoline	264	30	1–5	50	48	8/11	0.12 (0.03, 0.22)	37	36
	diesel	264	29	1–5	62	58	6/11	0.05 (−0.01, 0.12)	14	35
3	gasoline	2427	55	5–5	84	93	11/11	0.17 (0.11, 0.22)	65	37
	diesel	2427	54	4–5	81	89	11/11	0.15 (0.10, 0.18)	63	37

5.3 | Results

Table 1 summarizes, setting by setting, the 66 estimated models, one for each setting, fuel and city ($3 \times 2 \times 11$). Figure 3 gives the answer to Q1 city by city, Figure 5 depicts the size of the two pass-throughs, and Figure 4 two examples of spatial partitions induced by the SC-STEM model. See Application Supplement C for every model and every regime.

Local leadership (Q1).

- *Among the major brands, the nearest competitor leads almost everywhere.* In setting 3 the test rejects in 84% of the regimes for gasoline and in 81% for diesel. These regimes cover 93% and 89% of the pumps, and they are found in every city. The median long-run pass-through is 0.17 for gasoline and 0.15 for diesel: when the relative price of the nearest competitor moves permanently by one cent, that of the pump moves by about a sixth of a cent. The bootstrap interval excludes zero in about two regimes out of three.

- *Independent stations lead the major brands less.* In setting 1 the test rejects in 62% and 73% of the regimes, and the pass-through is about half as large, with medians of 0.08 and 0.05. The weakest case is Bari, where the nearest independent station is 5 to 8 kilometers away and leads only in small regimes, which cover 8% and 22% of the pumps.

- *Major brands lead independent stations unevenly.* In setting 2 the test rejects in half to two thirds of the regimes. All the independent stations of Florence, Venice and Messina follow their nearest major brand, but only half or fewer of those of Rome, Naples and Catania do. The samples are small, and the cities differ more than in the other settings.

- *Regimes with an Eni majority follow their neighbor less often.* Regimes in which most stations are Eni follow their neighbor in 72% of the cases in setting 3 and in 50% in setting 1, against 86% and 73% for the other regimes, and their pass-through

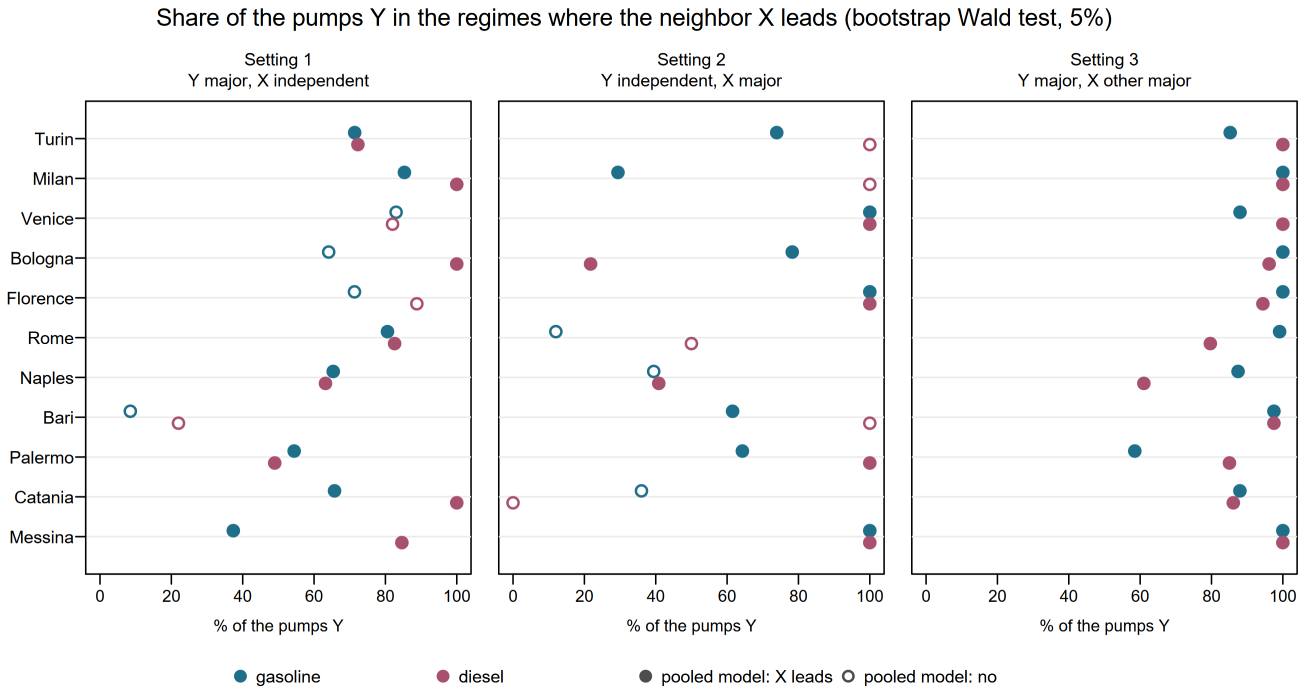


FIGURE 3 Local leadership by city: share of the pumps Y in the regimes where the bootstrap Wald test of the eight lags of the neighbor X rejects at 5%, for gasoline (blue) and diesel (red). Filled marks: the same test on the pooled STEM model also rejects; open marks: it does not. In setting 2 Florence and Venice, with fewer than ten independent stations, are fitted by the pooled model.

is smaller (see Section C4 of Application Supplement C). This finding should be interpreted as a difference in predictive dependence, not in conduct. A weaker response to the nearest competitor is consistent with prices set centrally by the brand, with different contracts between the brand and the station operators, or with leadership running in the opposite direction, which the settings do not separate by brand. Without costs and margins, it says nothing about market power.

- *The pooled model answers a different question.* It finds leadership in every model of setting 3. In setting 1, however, it misses leadership in Venice and Florence for both fuels and in Bologna for gasoline, although most pumps there are in regimes that follow their neighbor. In setting 2 it misses leadership for diesel in Turin, Milan, Rome and Bari. Where the regimes differ, the pooled model averages relations of different form and size, and the average may not be significant. Conversely, in five models the pooled model attributes to the whole city a leadership that holds only in regimes covering less than half of the pumps.

- *Station by station, the evidence is diluted.* The HC1 test rejects for 31 to 37% of the pumps, more often for close pairs: in setting 3, for 42% of the pumps within 500 meters of their neighbor against 31% between two and ten kilometers. A single station rarely reveals what its regime shows. The classical F test rejects for 51 to 60% of the pumps, an over-rejection due to the strongly heteroskedastic changes of sticky prices (see Section C5 of Application Supplement C).

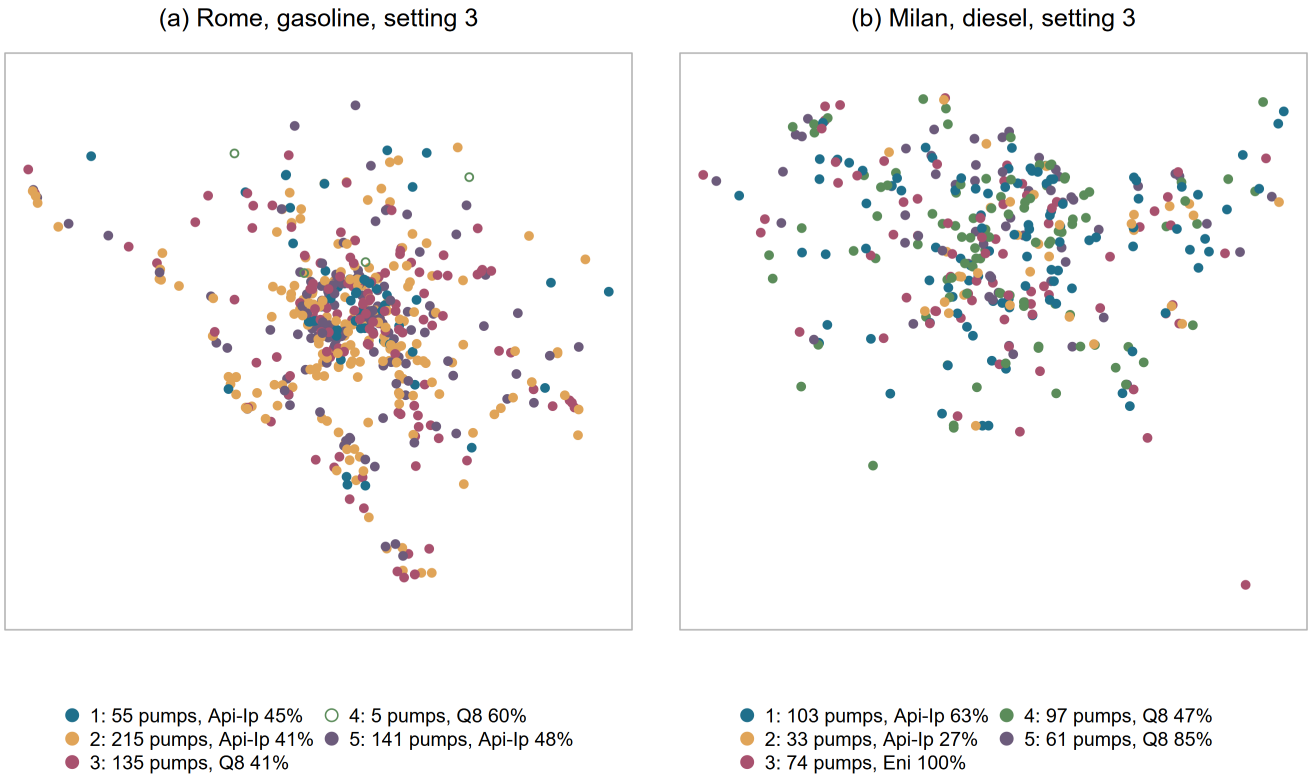


FIGURE 4 Examples of spatial partitions induced by the model for setting 3. Every pump Y is colored by its regime. Filled marks: the regime where the neighbor leads, open marks: where it does not. The legend gives the size of each regime and its most frequent brand with its share. Panel (a) Rome, gasoline: five regimes, four of them following their neighbor. Panel (b) Milan, diesel: regime 3 consists of a single brand (the 74 Eni stations of the city).

The regimes (Q2).

- *Many groups.* In settings 1 and 3 every model has four or five regimes, and 38 of the 44 stop at the cap of five, so the BIC would take more: a city's major-brand stations do not price as one group. With the independents of setting 2, 6 to 71 per city, $\hat{K}$ ranges from one to five.

- *Groups of brands, not areas.* The rule chooses $\hat{\phi} = 0$ in 29 models and $\hat{\phi} \leq 0.2$ in all but two, and the regimes interpenetrate in space (Figure 4). They are pricing groups: the adjusted Rand index[59] between regimes and brands is 0.07 to 0.48 (median 0.23), a moderate association, but well above the index between the regimes and a purely spatial partition with as many groups (k-means on the coordinates of the pumps), which is −0.02 to 0.10 (median 0.01) and lower in all the 44 models of settings 1 and 3. In 25 of these models one regime consists of Eni stations only.

The pass-through of the excise duties (Q3).

- *Large changes pass through unevenly.* In the week of the temporary cut of March 2022, 71% of the regimes for gasoline and 81% for diesel deviate from the national price, from about 6 cents below to 13 to 16 cents above (10th and 90th percentiles),

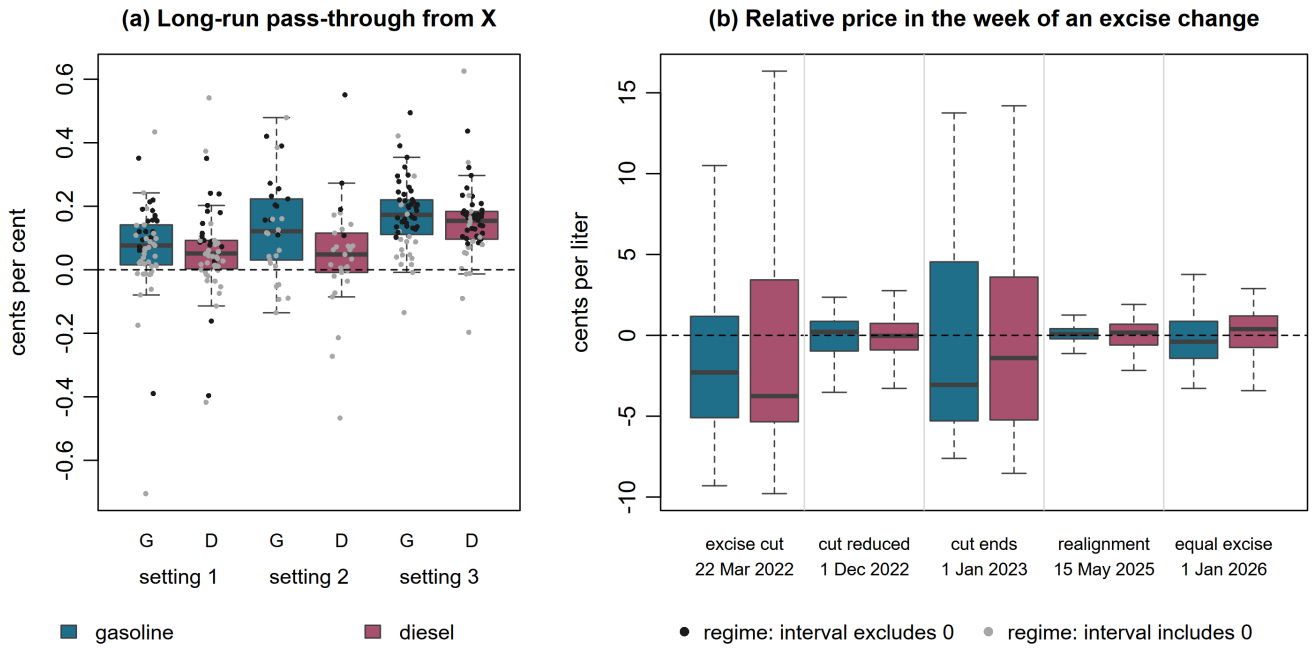


FIGURE 5 Panel (a): estimated long-run pass-through $\hat{\lambda}_k$ from the neighbor X by setting and fuel (boxplot of one value per regime); black points: the bootstrap 95% interval excludes zero. Panel (b): estimated coefficient of the pulse of the week of each change of the excise duties across the models (boxplot of one value per regime), that is, the estimated deviation, in cents per liter, of the relative price of the regime in that week.

and the regimes of one city differ by a median of 25 to 27 cents: some groups of stations passed the cut on within the week, others had not. The end of the cut, in January 2023, shows the same picture (Figure 5 b). A heterogeneous pass-through of a fuel tax discount has been documented for Germany by Frondel et al.[60]; here the groups that pass it on differently are found by the model rather than defined in advance.

- *Small changes pass through uniformly.* The realignment of May 2025, one and a half cents per liter in opposite directions on the two fuels, moves almost no regime away from the national price (1%). The realignment of the duties of January 2026 and the reduction of the cut in December 2022 move few.

5.4 | Implications

The results suggest three points of reflection: on local competition, on the geography of the Italian fuel industry, and on fiscal policy.

Local leadership is the rule among the major brands of Italian cities: the past prices of the nearest competitor of another brand help to predict the price of a station in almost every regime. This agrees with the evidence on single Italian markets, where stations react to the prices of competitors located within about one kilometer[61,62], and with the price leadership documented in other countries[63,47]. This leadership is predictive, in the sense of Granger, that is, the past prices of the competitor carry

information on the future price of the station, but the test does not say why. It is not by itself evidence of coordination or of market power, and we observe neither costs nor margins. It identifies, however, the channel through which a price change spreads in a local market, which matters to the firms of the industry as much as to competition authorities. Notably, since 2023, Italian stations must also display the regional average price published by the Ministry, which makes prices easier to compare and may ease their alignment where local competition is weak[51]; our window spans this change, and the coefficients of a regime average over the periods before and after it. Whether leadership sustains higher margins, as leadership, coordination and the speed of reaction to competitors do in other markets[64,47,65], requires data on margins. The groups that differ in leadership are associated with brands more than with districts. An analysis at the city level, or by brands fixed in advance, averages over groups that react differently, and may conclude that a city has no leader where most of its stations follow one, or the reverse.

Leadership is weaker in the South. Pooling the two fuels, the regimes where the neighbor leads cover 83% and 81% of the pumps of setting 1 in the North (Turin, Milan, Venice, Bologna) and in the Center (Florence, Rome), but only 51% in the South and the islands (Naples, Bari, Palermo, Catania, Messina); in setting 3 the shares are 96%, 91% and 85%. The weakest cases are Bari, Messina for gasoline and Palermo in setting 1, and Palermo for gasoline and Naples for diesel in setting 3. Distance does not explain the gap: the nearest neighbor is about as far in the North as in the South, with medians of about 4 kilometers in setting 1 and one kilometer in setting 3. Prices are also stickier in Palermo and Naples, where the Sunday price is most often unchanged from the previous week (see Section C1 of Application Supplement C). Where the nearest competitor leads, the pass-through is similar across the areas, with a median between 0.15 and 0.17 in setting 3. The response to the 2022 cut of the excise duties varies within cities rather than between areas. The median spread between the regimes of a city is 20 to 24 cents in each of the three areas; it is largest in Bari (34 cents), Palermo (27), Venice (26) and Milan (25), and smallest in Messina (11) and Turin (12). To our knowledge, local leadership has not been compared across Italian regions before. Studies of single markets find that territorial factors, and not only the number and distance of competitors, shape the prices of Italian stations[62,66], and the response of prices to the display of the regional average price lasted longer in the South and the islands than in the North, where competition is stronger and the network of stations denser[51]. Both findings are consistent with a gap between areas that distance does not explain.

Finally, a large and temporary cut of the excise duties reaches consumers at different speeds within the same city. In the week of the 2022 cut, the regimes of a city differed by about 25 cents per liter in their relative price, and some groups of stations had not yet passed the cut on. Small changes, such as the realignment of 2025, raised no such issue. Studies of the 2022 cuts find that, on average, they were largely passed on to consumers in Germany, France and Italy[67,68], as fuel taxes generally are[69]. Our results show that a complete average pass-through can hide groups of stations that lag behind, as Frondel et al.[60] found over space and time for the German fuel tax discount. The evaluation of a tax cut should therefore look at groups of stations, and the regimes identify them from the prices alone, without assumptions on brands or locations.

6 | CONCLUSION

We have introduced the SC-STEM model, which partitions a network of georeferenced locations, each observed over time, into a small number of latent spatial regimes, and gives each regime a complete spatio-temporal parameter set: its own regression coefficients, its own latent dynamics, and its own split of the residual variance between a local and a spatially structured component. The partition is estimated jointly with the parameters by maximizing a Potts-penalized likelihood, alternating an EM step within regimes with a sequential update of the labels. The construction borrows its algorithmic apparatus from spatially clustered regression for cross-sectional data[24] and from its extensions to spatial autoregressive, panel and small area models[34,35,36], and extends it to a spatio-temporal geostatistical design.

The simulations show that the procedure is reliable where it matters most: it rarely split a homogeneous network, it recovered well-separated regimes in nearly every replication from one hundred locations, and, with the right number of regimes, its estimates were centered on the true values. Weakly separated regimes need long series or many locations, and the selection rule merges them when the data cannot tell them apart. Our application to the retail fuel industry in Italy shows that local leadership, in the sense of Granger, is pervasive among the major brands of eleven Italian cities: the regimes that differ in it are associated with brands far more than with districts, and a model without regimes misses leadership where the regimes differ. The application measures how well the prices of the competitors predict those of a station, not market power, which would require data on costs and margins. The model also identifies, without defining them in advance, the groups of stations that passed the 2022 cut of the excise duties on more slowly.

SC-STEM inherits the assumptions of STEM: the response is univariate and Gaussian; the spatial covariance is exponential, hence isotropic and stationary within a regime; the network is fixed and observed at regular times; the covariates and the coordinates must be complete; and each regime must contain enough locations to support its own fit. In the application, the Gaussian assumption and the cap on the number of regimes qualify the evidence. Sticky prices give residuals with heavy tails and strong conditional heteroskedasticity (median kurtosis of 10 to 11), so the Gaussian model and its parametric bootstrap, with $B = 100$ draws, are approximations, and the size of the tests of leadership is not known; the cap of five regimes binds in 38 of the 44 models of settings 1 and 3, so a regime may merge groups of stations that behave differently (Sections C3 and C6 of Application Supplement C). Each of these limits points to a development of the model. Multivariate[70], functional[71] or non-Gaussian responses[72,73,74], gaps in the covariates, and a number of regimes estimated within the model, for instance through a prior on partitions[75,76], would widen its scope. For the tests of leadership, a bootstrap robust to time-varying variance, as proposed for Granger tests in vector autoregressions[77], and tests in the quantiles of the distribution, which in energy markets locate the ranges of the distribution where causality arises[78], would address the heavy tails and the heteroskedasticity of the prices.

Moreover, an assignment score based on the conditional density of a location given its neighbors, in the spirit of a Gaussian conditional autoregression [79,80], would use the spatial covariance that the present score discards. On the applied side, the approach suits any network of georeferenced series whose behavior may change across space: other retail and energy markets, labor and housing markets observed at a fine geographic scale, and environmental networks, where regimes with their own spatial covariance may improve interpolation and prediction at unmonitored sites [8,74,81]. More broadly, georeferenced series of this kind are a growing share of the data that data science handles in industry, from sensor networks to retail chains, and SC-STEM offers an interpretable way to find the groups of locations that behave alike within them.

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Conflict of interest

The author is affiliated with Fondazione Eni Enrico Mattei (FEEM), a non-profit research foundation established by Eni. Eni had no role in the design of the study, in the collection, analysis and interpretation of the data, in the writing of the manuscript or in the decision to submit it. The author declares no other potential conflict of interest.

Data availability statement

The fuel prices are published daily, station by station, by the Italian Ministry of Enterprises and Made in Italy through the Osservaprezzi Carburanti platform[49]. The data used in the application, namely the weekly prices of the 3,136 stations with their location and brand and the national mean prices, are available together with the code that reproduces all the results in the folder `papers/sc-stem-local-leader` of the GitHub repository of the `Stem` package (https://github.com/PaoloMaranzano/Stem).

Code availability

The methods of this paper are implemented in the `Stem` package for R[82,83], available at https://github.com/PaoloMaranzano/Stem together with a code supplement that documents the implementation. The code that reproduces the simulation study and the application is in the folder `papers/sc-stem-local-leader` of the same repository.

SUPPORTING INFORMATION

Additional information for this article is available online. Computational and Algorithmic Supplement A: the estimation algorithm, its accelerations and the computational details, with their derivations and numerical experiments. Simulation Supplement B: the design and the complete results of the simulation study. Application Supplement C: more details on the Italian retail fuel data and the results of every model and every regime.

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APPENDIX

A THE REFIT-WITH-CLUSTERING BOOTSTRAP

Algorithm 2 states the refit-with-clustering parametric bootstrap of Section 3.3.

Algorithm 2 Refit-with-clustering parametric bootstrap for SC-STEM

Input: data $\{z_{it}, x_{it}\}$, coordinates $s_1, \dots, s_d$, loading matrix A, selected $(\hat{K}, \hat{\phi})$, number of draws B.
Run Algorithm 1 at $(\hat{K}, \hat{\phi})$ on the data: partition $\hat{\mathcal{P}} = \{\hat{C}_1, \dots, \hat{C}_{\hat{K}}\}$ and estimates $\hat{\Psi}_1, \dots, \hat{\Psi}_{\hat{K}}$.
for $b = 1, \dots, B$ **do**
    for $k = 1, \dots, \hat{K}$ **do**
        Simulate the latent path $y^{*(k,b)}_{0:T}$ from Equations (2)–(3) at $\hat{\Psi}_k$.
        Simulate the errors $e^{*(k,b)}_t \sim N(0, \hat{\Sigma}_{e,k})$, $t = 1, \dots, T$, on the locations of $\hat{C}_k$, with their coordinates.
        Generate $z^{*(b)}_{it} = x_{it}^{\top}\hat{\beta}_k + A_i\, y^{*(k,b)}_t + e^{*(k,b)}_{it}$ for $i \in \hat{C}_k$, with the observed covariates.
    end for
    Restore the original order of the locations, so that every location keeps its neighbors in the graph, and remove the values that are missing in the data.
    Run Algorithm 1 at $(\hat{K}, \hat{\phi})$, with the same settings, on $z^{*(b)}$: partition $\hat{\mathcal{P}}^{(b)}$ and estimates $\hat{\Psi}^{(b)}_1, \dots, \hat{\Psi}^{(b)}_{\hat{K}}$.
    Map each regime of $\hat{\mathcal{P}}^{(b)}$ onto the regime of $\hat{\mathcal{P}}$ with which it shares most locations, resolving conflicts by decreasing overlap; discard the draw if a regime is not recovered.
    Store the aligned estimates $\hat{\Psi}^{(b)}_1, \dots, \hat{\Psi}^{(b)}_{\hat{K}}$ and the partition $\hat{\mathcal{P}}^{(b)}$.
end for
return the aligned draws of the regime-specific parameters; their standard errors; the normal, basic, percentile and bias-corrected confidence intervals; the percentile tests of the differences between regimes; the share of usable draws; the stability of the partition across the draws.

B EXAMPLES OF SIMULATED DATA

Figure B1 shows one replication of the design of Section 4 where the regimes lie in the four geometries and with unbalanced sizes, and the series of every scenario-variant at the reference cell. The regimes of S1s-`ind` move independently of each other. In the `shr` variants they share the latent shocks and differ by level, amplitude and persistence, by much in S1s-`shr` and by little in S1w-`shr`; with common regression coefficients (S3beta-`shr`) their mean levels coincide.

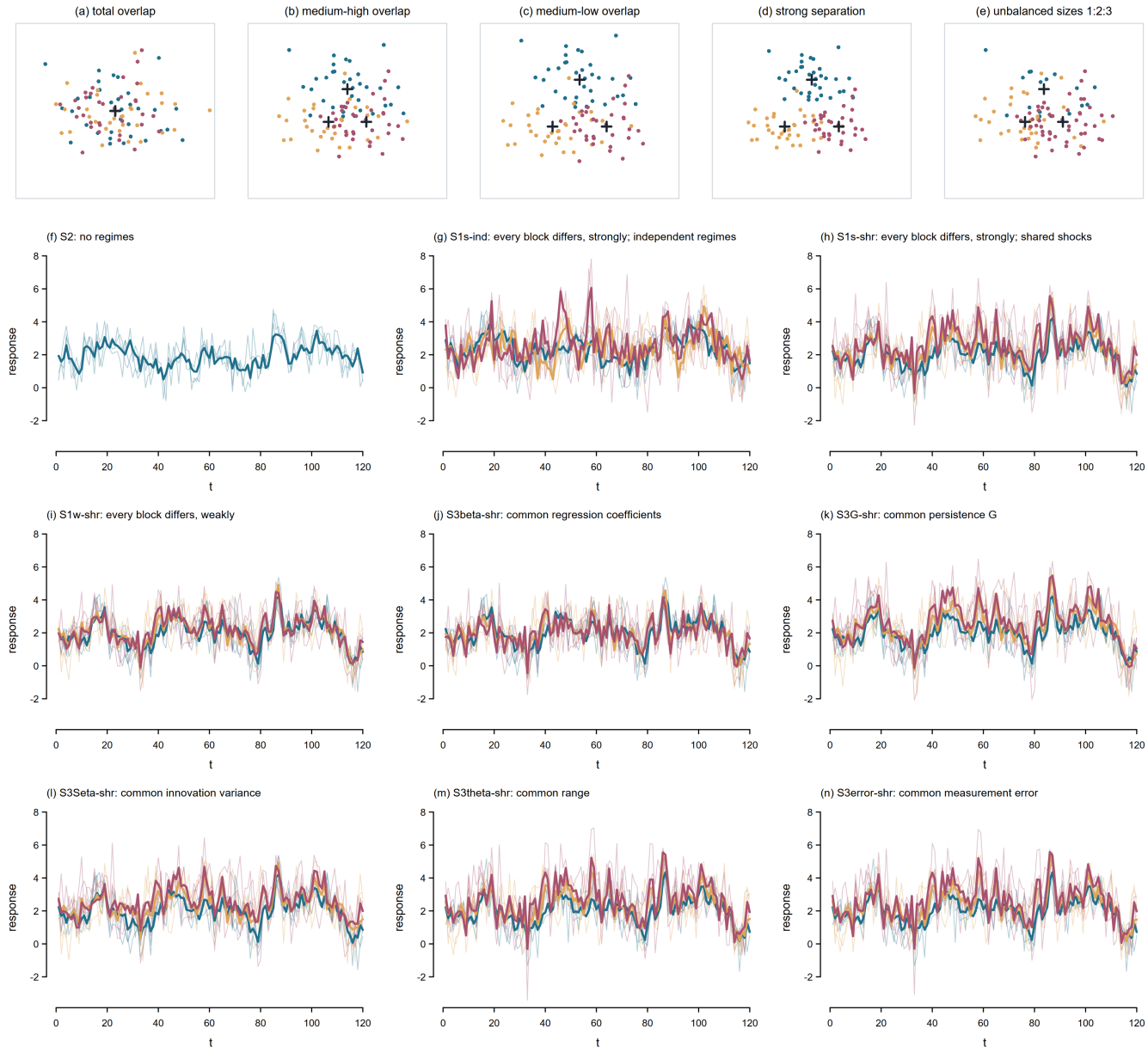


FIGURE B1 The simulation design at $d = 100$ and $T = 120$, first replication. (a)–(e) The locations, colored by regime, in the four geometries and with regime sizes 1 : 2 : 3; the crosses mark the centers of the regimes. (f)–(n) The response of every scenario-variant: three locations of each regime (thin lines) and the mean of the response over the locations of each regime (thick lines).

C FROM POSTED PRICES TO RELATIVE PRICES

Figure C1 shows the two transformations of Section 5.1 for two gasoline stations of Rome, an Eni station and an independent one, chosen among those with ordinary dynamics and with different premiums. The posted Sunday prices (Panel a) share the stochastic trend of costs and taxes. Subtracting the national mean of the week removes this trend (Panel b): the two stations then fluctuate around their own premiums, 0.8 cents above the national price for the Eni station and 2.7 cents below it for the

independent one. Subtracting the mean of each station removes the premiums (Panel c) and gives the relative prices on which the models are fitted.

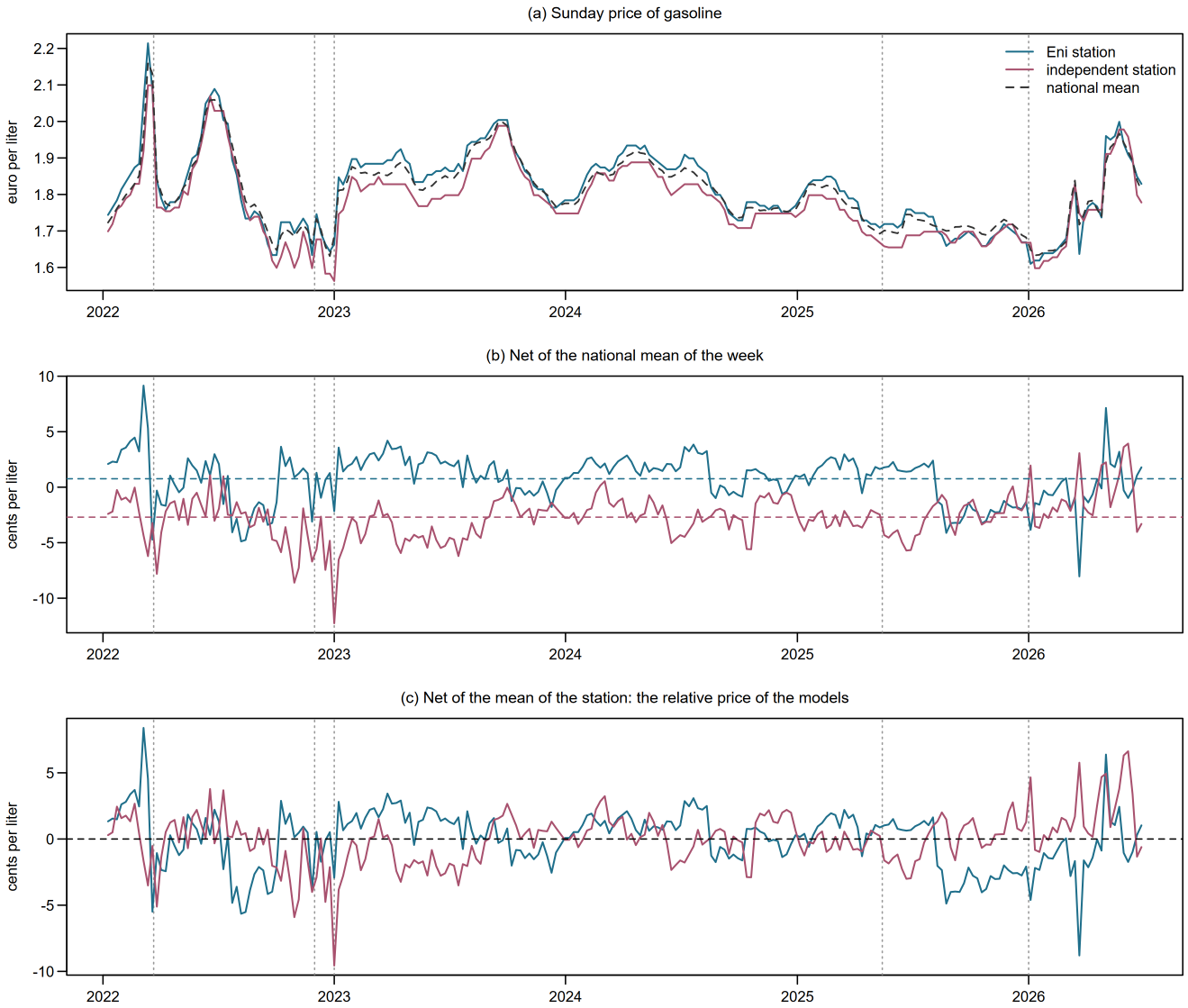


FIGURE C1 Two gasoline stations of Rome, 2022–2026. (a) Sunday price, with the national mean of the week (dashed). (b) Price net of the national mean, with the mean of each station (dashed). (c) Price net of the national mean and of the mean of the station: the relative price of the models. Dotted vertical lines: the changes of the excise duties.

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